

Darren Trieu

<https://d1trieu123.github.io/> - darrentrieu@gmail.com - (858) 429-8209

Education

University of California, San Diego | *M.S Computer Science GPA 3.79* Sept. 2023 - Dec. 2024

- Coursework : Probabilistic Reasoning & Learning, Databases, Programming Languages, Algorithm Design and Analysis, Software Engineering, Networking and Distributed Programming

University of California, San Diego | *B.S Pharmacological Chemistry GPA 3.77* Sept. 2018 - Jun. 2022

Work Experience

Summer Programming Intern - UCSD Weibel Lab June 2024 - Present

- Core contributor to the LifeSaver project, particularly in developing the AILA server, data from various medical devices and interacts with users to provide medical diagnoses and consultations.
- Manage the TCP socket using Node.js, Socket.IO, and Express to handle device input, process data, and send configuration data and messages back to clients.
- Develop RESTful endpoints for processing audio data, integrating Deepgram for real-time transcription and OpenAI for natural language processing.
- Design algorithms leveraging Bayesian networks, information theory, and greedy algorithms to diagnose patients effectively.
- Participate in scrum meetings, providing project updates and collaborating with team members to ensure project alignment and progress.

Solution Maker - UCSD Kadonaga Lab May 2019 - Jun. 2021

- Used WordPress to develop and manage the laboratory website to centralize lab research information.
- Managed the laboratory stock of crucial acids, bases, and bacteria to keep laboratory flow consistent.

Projects

SurfStore Cloud Storage Project- Go, gRPC, Protocol Buffers Mar. 2024 - Jun. 2024

- Designed and implemented a cloud file storage service that supported file synchronization and retrieval.
- Implemented a scalable BlockStore utilizing consistent hashing for efficient data distribution & retrieval of data blocks between various files.
- Implemented a scalable MetaStore that utilized the RAFT consensus algorithm to ensure file metadata consistency and fault tolerance across multiple servers.
- Developed the interaction algorithm between stores for efficient storage and retrieval of data.

UCal San Diego - Javascript, React Native, CSS, MongoDB, Git Jan. 2024 - Mar. 2024

- Implemented a mobile application that can be utilized to track food, nutrient, and health information.
- Utilized Agile practices to manage workflow, deliverables, and backlogs to plan and distribute labor.
- Used React Native to create the front end that interacted with a MongoDB database using a Netlify API.

XQuery Interpreter - Java, BASH Jan. 2024 - Mar. 2024

- Created a XQuery interpreter using ANTLR4 to create parse trees given XQuery expressions.
- Developed algorithm to navigate parse trees and search XML documents to return answers to queries.
- Implemented a query rewriter that detects implicit joins and outputs an explicit join query to allow for hash-based joining, avoiding naive query interpretation.

Terminal Asteroid Typing Game - Haskell, Git Oct. 2023 - Dec. 2023

- Lead a team of 4 to create a terminal-based game using functional programming and Haskell.
- Utilized the Brick Library employing widgets and attributes to render graphics in the terminal.
- Manipulated Haskell fundamentals such as state, functors, and monads to manage game features.

Skills

Languages: Python, Java, SQL, Javascript, CSS, HTML, Go

Technologies: React, MySQL, Pandas, ANTLR4, BASH, MongoDB, GRPC, Protocol Buffers, Node JS

Workflow: Git, Github, Agile